RiskRadar

Financial NLP and portfolio scenarios

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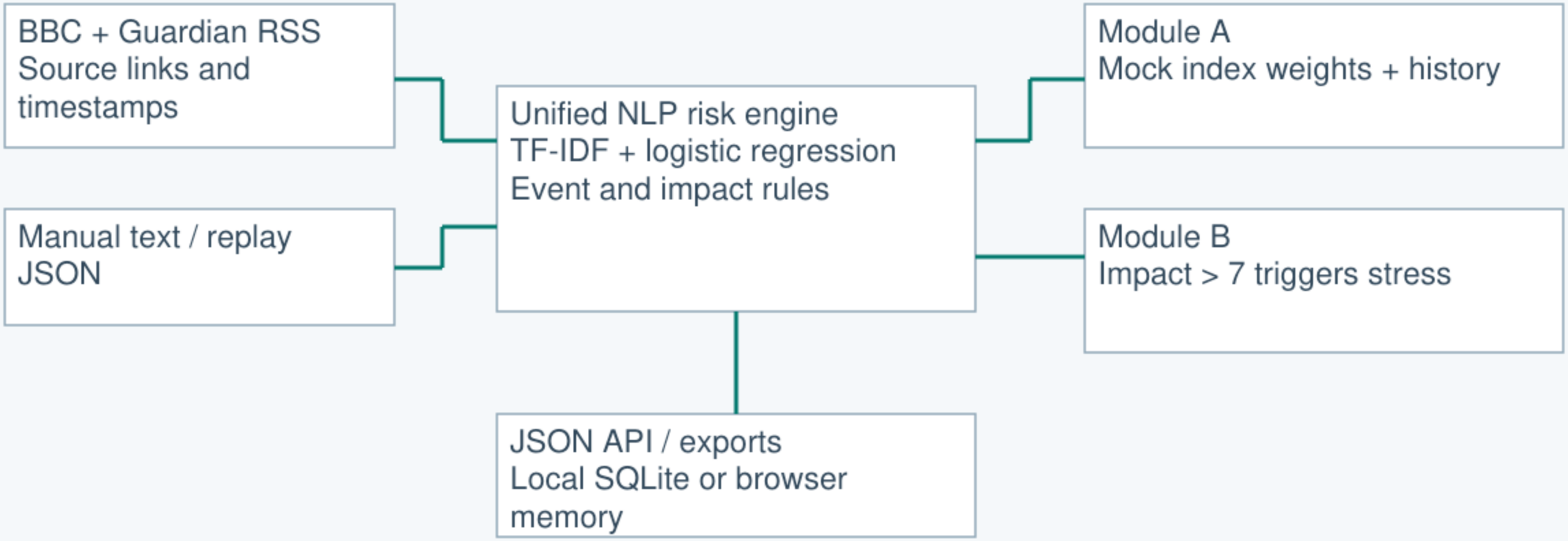
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Problem and approach

Financial news arrives as unstructured text from multiple sources
Analysts must connect an event with the exposures it could affect

One engine emits sentiment, event category and severity evidence
Two consumers demonstrate different decisions:
- Module A changes sentiment weights in a mock stock index
- Module B simulates asset shocks after a high-impact event

System design



Input records feed the engine, then both modules consume its structured output.

Implementation and sentiment results

Training: financial tweets from the public zeroshot dataset
9,374 training records and 2,318 evaluation records after cleaning

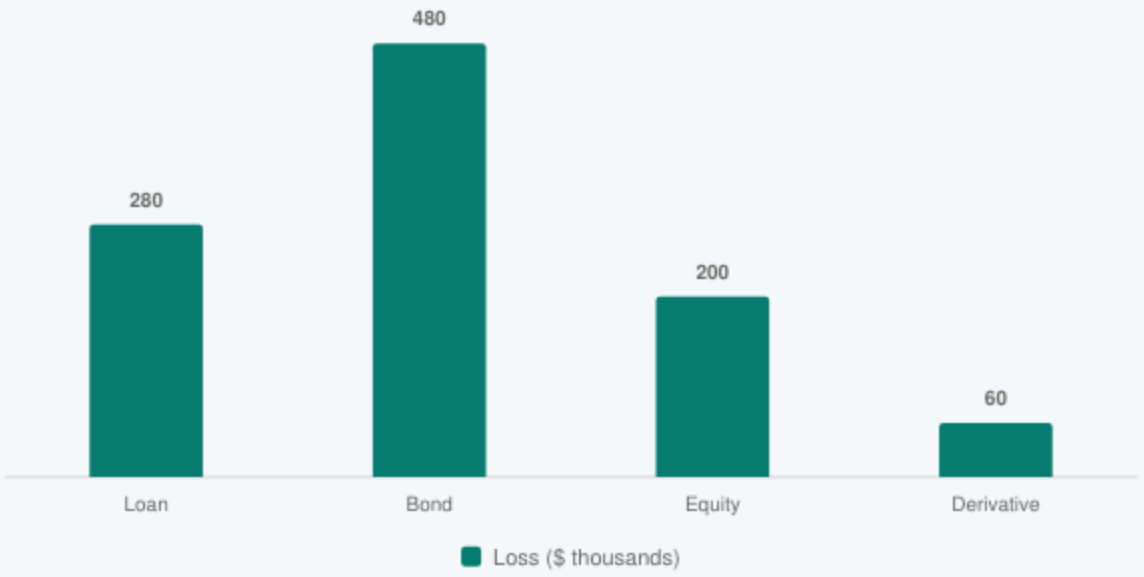
Sentiment accuracy 82.05% Macro F1 0.761
Exact duplicates, label conflicts and split overlap removed

Portable Python/JS inference and JSON outputs
Event category and impact remain transparent keyword heuristics

Module B: portfolio stress test

Baseline: $10.00 million
After stress: $8.98 million
Loss: $1.02 million (10.2%)

War replay: impact 9 > 7
Every run starts from baseline



Assumed equity shock -10%, rate shock +2 pp, credit-spread shock +1.5 pp

Module A: mock index rebalancing

Ten equal starting positions, then normalized sentiment weights
Raw weight = 1 + 0.5 × latest company sentiment
History and JSON export make changing allocations inspectable



Domain impact, limits and next steps

News triage: one inspectable format preserves source and evidence
Stress view: relate event assumptions to asset-level losses
Mock index: demonstrate a second consumer of the same signal

Limits: simple rules, sentiment domain shift and linear valuation
Next: temporal evaluation and market-calibrated severity/shocks

Public code, deck and data accompany the working dashboard